ANMOL PAHWA

Phone: +91 44 2257 4275
E-mail: anmpahwa@iitm.ac.in
Website: lognitilab.github.io
URL: civil.iitm.ac.in/faculty/anmpahwa

338
Building Sciences Block
IIT Madras, Chennai
600036

EDUCATION

Ph.D. University of California, Davis
September 2017 – June 2023 Civil and Environmental Engineering
Dr. Miguel Jaller *Assessing Sustainability of E-commerce Goods Distribution.*

B.Tech. Indian Institute of Technology – Delhi
August 2013 – May 2017 Civil Engineering
Dr. Geetam Tiwari *Indicators of Reliability and Variability in Bus Systems.*

WORK EXPERIENCE

Assistant Professor Indian Institute of Technology – Madras
July 2024 – Present Civil Engineering

Postdoctoral Scholar Institute of Transportation Studies
July 2023 – June 2024 University of California, Davis

PUBLICATIONS

Journal Publications

Kirubakaran, B., & Pahwa, A. (Under Review) Evolving Consumer Shopping Behavior Patterns in India: Evidence from the Indian Time Use Survey. *Transportation Research Part A: Policy and Practice*

Xiao, R., Jaller, M., Pahwa, A. & Cunha, C.B. (2026). Route-informed planning for medium- and heavy-duty truck charging infrastructure in California. *Transportation Research Part D: Transport and Environment*, 161, 105631.

Kirubakaran, B., Pahwa, A., Kancharla, S. R., & Waller, S. T. (2026). A holistic continuous approximation framework for strategic last-mile distribution planning. *Transportation Research Interdisciplinary Perspectives*, 35, 101720.

Dennis-Bauer, S., Pahwa, A., & Jaller, M. (2026). Explicit consideration of human exposure to minimize freight routing impacts. *Transportation Research Part D: Transport and Environment*, *151*, 105133.

Pahwa, A., & Jaller, M. (2025). Assessing the sustainability of last-mile distribution strategies to manage expedited shipping with dynamic and stochastic demand. *Transportation Research Part E: Logistics and Transportation Review*, 201, 104273.

Pahwa, A., & Jaller, M. (2024). Evaluating private and system-wide impacts of freight eco-routing. *Transportation Research Part D: Transport and Environment*, 130, 104170.

Jaller, M., Xiao, R. I., Dennis-Bauer, S., Rivera-Royero, D., & Pahwa, A. (2023). Estimating last-mile deliveries and shopping travel emissions by 2050. *Transportation Research Part D: Transport and Environment*, 123, 103913.

Pahwa, A., & Jaller, M. (2023). Assessing last-mile distribution resilience under demand disruptions. *Transportation Research Part E: Logistics and Transportation Review*, 172, 103066.

Pahwa, A., & Jaller, M. (2022). A cost-based comparative analysis of different last-mile strategies for e-commerce delivery. *Transportation Research Part E: Logistics and Transportation Review*, 164, 102783.

Jaller, M., Otero-Palencia, C., & Pahwa, A. (2020). Automation, electrification, and shared mobility in urban freight: opportunities and challenges. *Transportation Research Procedia*, 46, 13-20.

Jaller, M., & Pahwa, A. (2020). Evaluating the environmental impacts of online shopping: A behavioral and transportation approach. *Transportation Research Part D: Transport and Environment*, 80, 102223.

Pahwa, A., Lakhotia, S., & Tiwari, G. (2017). Methodology for using GPS data from buses to assess link-based travel time variation. *Journal of the Eastern Asia Society for Transportation Studies*, 12, 1238-1251.

Book Chapters

Blessy, K., & Pahwa, A. (2026). Opportunities and Challenges in Adopting Alternate Fuel Technologies for Sustainable Long-Haul Freight Transportation. *Smart Villages and Sustainable Rural Development. World Scientific.*

Jaller, M., Pahwa, A., Otero-Palencia, C., & Pourrahmani, E. (2023). Overview of city logistics and urban freight transport operations. *Handbook on City Logistics and Urban Freight. Edward Elgar Publishing*

White Papers

Jaller, M., Pahwa, A., Saphores, J. D., & Hyland, M. (2024). Benefits, Challenges, and Opportunities of Different Last-Mile Delivery Strategies. *University of California Institute of Transportation Studies.*

Jaller, M., Pahwa, A., & Center, M. T. (2023). Sketch Planning Tool for Sustainable and Resilient Urban Goods Distribution: User Manual. *METRANS Transportation Center (Calif.).*

Jaller, M., & Pahwa, A. (2023). Coping with the Rise of E-commerce Generated Home Deliveries through Innovative Last-Mile Technologies and Strategies. *National Center for Sustainable Transportation.*

Giuliano, G., Boarnet, M., Jaller, M., Franco, W., Binder, R. B., Dexter, S., Fang, J., David Flores, M., Pahwa, A., Marley, R., & Wang, B. (2022). Improving Environmental Justice and Mobility in Southeast Los Angeles. *Pacific Southwest Region University Transportation Center (UTC).*

Jaller, M., & Pahwa, A. (2022). Assessing E-retailers' Resilience During the COVID-19 Pandemic. *Institute of Transportation Studies, UC Davis.*

Jaller, M., Xiao, R., Dennis, S., Rivera-Royero, D., & Pahwa, A. (2022). National Impacts of E-commerce Growth: Development of a Spatial Demand Based Tool. *National Center for Sustainable Transportation.*

Jaller, M., Pahwa, A., & Zhang, M. (2021). Cargo Routing and Disadvantaged Communities. *Pacific Southwest Region University Transportation Center.*

Jaller, M., & Pahwa, A. (2020). Analytical Modeling Framework to Assess the Economic and Environmental Impacts of Residential Deliveries and Evaluate Sustainable Last-Mile Strategies. *National Center for Sustainable Transportation.*

Rodier, C., Jaller, M., Pourrahmani, E., Bischoff, J., Freedman, J., & Pahwa, A. (2018). Automated vehicle scenarios: Simulation of system-level travel effects using agent-based demand and supply models in the San Francisco Bay area. *National Center for Sustainable Transportation.*

Policy Briefs

Jaller, M., & Pahwa, A. (2025). Last-Mile Delivery Innovations and Best Practices in the Age of E-commerce. *National Center for Sustainable Transportation.*

Jaller, M., & Pahwa, A. (2021). Minimizing the Impact of Freight Traffic on Disadvantaged Communities. *Institute of Transportation Studies, UC Davis.*

Jaller, M., & Pahwa, A. (2021). The Sustainability of Alternative Last-Mile Delivery Strategies. *National Center for Sustainable Transportation.*

PRESENTATIONS

Lectern Presentations

Pahwa, A. (2026). *Challenges and Opportunities in Long-Haul and Last-Mile Freight Transportation.* Urban Mobility and Logistics Training and Workshop, CiSTUP, IISc Bangalore, India.

Blessy, K., & Pahwa, A. (2025). *A Holistic Continuous-Approximation Framework for Strategic Last-Mile Distribution Planning.* Presented at the METRANS International Urban Freight Conference, Los Angeles, CA, USA.

Xiao, R., Jaller, M., Pahwa, A. & Cunha, C.B. (2025). *Data-Driven Approach for the Optimal Planning of Medium and Heavy-Duty Truck Charging Infrastructure.* Presented at the METRANS International Urban Freight Conference, Los Angeles, CA, USA.

Pahwa, A., & Jaller, M. (2024). *Navigating the Last-Mile: Opportunities and Challenges towards Sustainable E-commerce Deliveries.* Presented at the Transportation Research Board 103rd Annual Meeting, Washington, D.C.

Pahwa, A., & Jaller, M. (2022). *Freight eco-routing – carrier's trade-offs and system-wide impacts.* Presented at the METRANS International Urban Freight Conference, Los Angeles, CA, USA.

Pahwa, A., & Jaller, M. (2022). *Can Truck Eco-Routing Bridge the Gap in Transition to Zero-Emissions?* Presented at the Transportation Research Board 101st Annual Meeting, Washington, D.C.

Pahwa, A., Lakhotia, S., & Tiwari, G. (2017). *Methodology for using GPS data from buses to assess link-based travel time variation.* 12th International Conference of EASTS, Ho Chi Minh City, Vietnam.

Poster Presentations

Blessy, K., & Pahwa, A. (2026). *Assessing Pre- and Post- Pandemic Shopping Behavior in Indian Consumers.* To be Presented at World Conference on Transportation Research 2026, Toulouse, France.

Blessy, K., & Pahwa, A. (2026). *Assessing Distribution Structures of Conventional and Quick Commerce Across Low- and High- Income Markets.* To be Presented at World Conference on Transportation Research 2026, Toulouse, France.

Blessy, K., & Pahwa, A. (2025). *A Holistic Continuous-Approximation Framework for Strategic Last-Mile Distribution Planning.* Presented at the Conference of Transportation Research Group (CTRG), Guwahati, India.

Dennis, S., Pahwa, A., & Jaller, M. (2024). *Cargo Routing, Health Impacts, and Disadvantaged Communities.* Presented at the Transportation Research Board 103rd Annual Meeting, Washington, D.C.

Pahwa, A., & Jaller, M. (2020). *Evaluating costs and distribution structure in last-mile deliveries under short time-windows.* Presented at the Transportation Research Board 99th Annual Meeting, Washington, D.C.

Bischoff, J., Rodier, C. J., Pourrahmani, E., Jaller, M., Pahwa, A., & Maciejewski, M. (2019). *Competition among Automated Taxis, Transit, and Conventional Passenger Vehicles: Traffic Effects in the San Francisco Bay Area.* Presented at the Transportation Research Board 98th Annual Meeting, Washington, D.C.

Jaller, M., & Pahwa, A. (2019). *Evaluating the Environmental Impacts of Online Shopping: A Behavioral Analysis using the American Time Use Survey (ATUS) Data.* Presented at the Transportation Research Board 98th Annual Meeting, Washington, D.C.

RESEARCH GRANTS

Principal Investigator on *Network Economic Impact Assessment of India's High-Speed Rail Programme Using a Validated Multi-Regional Transport CGE Model*. Funded by National High-Speed Rail Corporation Limited (2026-27, ₹2.065Cr)

Principal Investigator on *A Digital Twin for Adaptive Bus Fleet Management – Enhancing Efficiency and Resilience under Disruptions*. Funded by Office of Global Engagement, IIT Madras (2026-27, ₹9.43L)

Principal Investigator on *Decision Support Tool for Sustainable Urban Freight Management in Bengaluru*. Funded by Directorate of Urban Land Transport, Govt. of Karnataka (2025-26, ₹20.82L)

Principal Investigator on *Developing Battery-Aware Operations for Autonomous and Unmanned Last-Mile Delivery Systems*. Funded by IC&SR IIT Madras (2025-27, ₹49.9L)

Co-Principal Investigator on *PACT-Digi-PULSE - Partnering for Advancing Carbon-neutral Transitions using Digital-twin based Planning for Urban Living and Sustainable Environment*. Funded by Palladium International Limited (2025-27, ₹5.6Cr)

Co-Principal Investigator on *Sustainable Logistics Infrastructure and Planning*. Funded by Fedex Corporate Services Inc. (2025-28, ₹25L)

HONORS AND AWARDS

Young Faculty Mobility Grant by Office of Global Engagement, IIT Madras (2026)
Dissertation Grant by National Center for Sustainable Transportation Research Grant (2021)
PhD Fellowship by National Center for Sustainable Transportation Research Grant (2017)

PROFESSIONAL SERVICE

Committee Member

Transportation Research Group of India – Lifetime Member
Transportation Research Board: Standing Committee on Urban Freight Transportation – Friend
World Conference on Transportation Research – Lifetime Member

Peer-Reviewer

Applied Sciences
Central European Journal of Operations Research
Civil Engineering and Environmental Systems
Cogent Engineering
Conference of Transportation Research Group of India
Computers and Industrial Engineering
IEEE Transactions on Transportation Electrification
International Journal of Environmental Research and Public Health
International Journal of Logistics
International Journal of Systems Science: Operations & Logistics
Journal of Advanced Transportation
Journal of The Institution of Engineers (India): Series A
Networks and Spatial Economics
NPJ Sustainable Mobility and Transport
Sustainable Futures
Sustainability
Systems
Transactions on Engineering Management
Transport Policy
Transportation in Developing Economies
Transportation Letters: The International Journal of Transportation Research
Transportation Planning and Implementation Methodologies for Developing Countries
Transportation Planning and Technology
Transportation Research Board Annual Meeting
Transportation Research Interdisciplinary Perspectives
Transportation Research Part A: Policy and Planning
Transportation Research Part B: Methodological
Transportation Research Part C: Emerging Technologies
Transportation Research Part D: Transport and Environment
Transportation Research Part E: Logistics and Transportation Review
Transportation Research Record

Trauma care
Urban Transitions
World Symposium on Transport and Land Use Research

Supervisor

Postdoc

Ph.D.

Blessy K. (2024-present)
Stephen Babu (2025-present)
Nagasai Panidarapu (2025-present)
Dayanidha S (2025-present)
Hana Aboobacker (2026-present)

M.S.

Leena Pramod Khandait (2025-present)
A Ranjith Kanna (2026-present)

M.Tech.

Varun A. (2024-25)
Harshvardhan Shivram Ket (2025-26)
Aditya Aggarwal (2025-26)

B.Tech.

Arvind Vidhyashankar (2025-26)
Om Ramavtar Yadav (2025-26)

Course Instructor

CV 2060 – Transportation Engineering
CV 5540 – Data Analysis and Computational Techniques for Transportation Engineers
CV 5972 – Optimisation Techniques in Transportation Engineering
CV 5810 – Urban Transportation Planning